

Notes: Sufi (RFS, 2009)

Bank Lines of Credit in Corporate Finance: An Empirical Analysis

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1 Big picture

- **Question.** Why do some firms manage liquidity with bank lines of credit while others rely on cash? The paper asks whether credit lines are true liquidity insurance, or whether their availability is itself conditional on firm fundamentals.
- **Core idea.** Lines of credit are not perfect substitutes for cash. They are useful liquidity buffers mainly for firms that can maintain strong cash flow and remain compliant with bank covenants.
- **Empirical setting.** The paper studies U.S. nonfinancial public firms from 1996 to 2003. It hand-collects detailed information on lines of credit from annual 10-K filings for a random sample of 300 firms and uses text searches to classify line access in a much larger Compustat sample.
- **Main result.** High cash flow predicts access to bank lines of credit and a larger share of liquidity provided by credit lines. Low-cash-flow firms rely more heavily on cash.
- **Mechanism.** Banks use cash-flow-based covenants. When cash flow falls and covenants are violated, banks restrict credit-line availability.
- **Financial constraints implication.** Lack of access to a line of credit is a powerful measure of financial constraints. Traditional constrained-firm categories, such as small firms or unrated firms, exhibit cash saving out of cash flow mainly when they also lack credit-line access.
- **Economic takeaway.** Bank lines of credit provide liquidity only for firms that are sufficiently healthy to preserve access. Cash remains valuable because it is unconditional liquidity.

2 Incremental contribution of the paper

- **Bridges two literatures.** The cash-holdings literature emphasizes internal liquidity because external finance is costly. The credit-line literature emphasizes banks as efficient providers of liquidity insurance. This paper directly compares the two liquidity tools.
- **New measurement contribution.** The paper builds one of the first large empirical data sets on public firms' lines of credit using annual 10-K filings. It observes whether a firm has a line, the size of the line, the used and unused portions, and covenant violation status.
- **Contractual mechanism.** The paper shows that the availability of credit lines depends on financial covenants, especially cash-flow-based covenants. This distinguishes actual credit-line contracts from the fully committed liquidity contracts often described in theory.
- **Liquidity-substitution contribution.** It shows that cash and credit lines are substitutes only for some firms. For low-cash-flow or high-volatility firms, cash is not redundant because bank liquidity can disappear exactly when the firm is weak.
- **Financial-constraints contribution.** It proposes line-of-credit access as a direct, contract-based measure of financing constraints and shows that it improves on traditional proxies such as payout, size, bond ratings, and commercial paper ratings.
- **Empirical corporate finance contribution.** The paper documents that lines of credit are large and widely used, including by firms with public debt access and even by firms with no balance-sheet debt outstanding.

3 Data and measurement

3.1 Full Compustat and 10-K search sample

- The full sample starts from nonfinancial U.S. Compustat firms with sufficient accounting data from 1996 to 2003.
- The final full sample contains 4,503 firms and 28,447 firm-year observations.
- The author links each firm-year to its annual 10-K filing and searches for terms such as “credit facility,” “revolving credit agreement,” “line of credit,” and related expressions.
- This produces an indicator for whether the firm has a bank line of credit.
- The paper manually checks common false-positive phrases such as “no line of credit” or “terminated our line of credit.”

Interpretation: The full sample is useful for studying the extensive margin: whether a firm has a line of credit at all. Its disadvantage is that text classification can generate measurement error.

3.2 Random hand-collected sample

- The random sample contains 300 firms and 1,908 firm-year observations.
- The author manually reads 10-K filings to collect total line of credit, used line of credit, unused line of credit, and covenant violation information.
- The hand-collected sample allows the paper to measure the intensive margin of line use, not only the existence of a line.
- It also allows the author to evaluate the search algorithm. In the random sample, the search method produces false positives and false negatives, but the hand-collected data provide a cleaner benchmark.

Interpretation: The random sample is smaller but richer. It is the core sample for the bank-liquidity-to-total-liquidity ratios and the covenant tests.

3.3 Main liquidity variables

- **Line of credit indicator:** equals one if the firm has access to a bank line of credit.
- **Total line of credit/assets:** the committed line scaled by assets.
- **Unused line/assets:** the undrawn amount available to the firm.
- **Used line/assets:** the amount already drawn.
- **Bank liquidity to total liquidity:**

$$\frac{\text{Line of credit}}{\text{Line of credit} + \text{Cash}}.$$

- **Unused bank liquidity to total liquidity:**

$$\frac{\text{Unused line of credit}}{\text{Unused line of credit} + \text{Cash}}.$$

Interpretation: The liquidity ratio is important because total liquidity demand can differ across firms. The ratio asks whether liquidity is supplied by banks or by internal cash.

3.4 Key firm characteristics

- Cash flow is measured mainly as EBITDA divided by non-cash assets.
- Other controls include asset tangibility, firm size, cash-adjusted net worth, cash-adjusted market-to-book, industry sales volatility, cash-flow volatility, index membership, exchange listing status, and firm age.
- The paper scales several variables by non-cash assets to avoid mechanical correlations caused by cash being part of the denominator.

Interpretation: The focus on non-cash assets is a careful measurement choice because cash and credit-line use are jointly determined.

4 Models and empirical specifications

4.1 Credit-line access: extensive margin

The first empirical model relates line-of-credit access to lagged firm characteristics:

$$\Pr(\text{Line}_{it} = 1) = \Phi(\alpha_t + \alpha_{\text{industry}} + X'_{i,t-1}\beta).$$

- The dependent variable equals one if firm i has a line of credit in year t .
- The main explanatory variable is lagged cash flow.
- The model is estimated as a probit and reports marginal effects.

Interpretation: A positive cash-flow coefficient means that firms with stronger internal performance are more likely to have bank liquidity available.

4.2 Bank liquidity share: intensive margin

For the random sample, the paper estimates OLS models of the form:

$$\frac{\text{Line}_{it}}{\text{Line}_{it} + \text{Cash}_{it}} = \alpha_t + \alpha_{\text{industry}} + X'_{i,t-1}\beta + \varepsilon_{it}.$$

It also estimates analogous models using unused lines in the numerator and restricts some specifications to firms that already have a line of credit.

Interpretation: This asks whether, conditional on overall liquidity, firms with higher cash flow tilt their liquidity management toward bank lines rather than cash.

4.3 Distress interaction

To test whether cash flow matters most when banks are worried about default risk, the paper interacts cash flow with a low-distress indicator:

$$\text{LineUse}_{it} = \alpha + \beta_1 CF_{i,t-1} + \beta_2 \text{LowDistress}_{i,t-1} + \beta_3 CF_{i,t-1} \times \text{LowDistress}_{i,t-1} + \Gamma' X_{i,t-1} + \varepsilon_{it}.$$

- Low distress is based on Altman's z -score relative to the sample median.
- The negative interaction means cash flow matters less for safer firms.

Interpretation: Banks care especially about current cash flow when the firm is financially fragile.

4.4 Covenant violation models

The paper estimates firm fixed effects models for covenant violations:

$$\text{Violation}_{it} = \alpha_i + \alpha_t + X'_{it}\beta + \varepsilon_{it}.$$

It then estimates how violation affects future credit-line availability:

$$\text{Line}_{it} = \alpha_i + \alpha_t + X'_{i,t-1}\beta + \text{hetaViolation}_{i,t-1} + \varepsilon_{it}.$$

Interpretation: The first equation identifies which firm conditions predict default on financial covenants. The second asks whether banks reduce credit-line availability after a covenant violation.

4.5 Cash-flow sensitivity of cash

Following Almeida, Campello, and Weisbach, the paper estimates:

$$\Delta \text{Cash}_{it} = \alpha_i + \alpha_t + \beta CF_{it} + \gamma Q_{it} + \delta \text{Size}_{it} + \varepsilon_{it}.$$

- A positive cash-flow sensitivity of cash is interpreted as evidence of financial constraints.
- The paper compares the credit-line access measure with traditional constraint measures.

Interpretation: The key test is whether firms classified as constrained by traditional measures still save cash out of cash flow once credit-line access is taken into account.

5 Identification and empirical design

5.1 What the design identifies

- The design is not a randomized experiment. It primarily identifies robust conditional associations between firm fundamentals, credit-line use, covenant violations, and cash policy.

- The strongest causal-looking evidence comes from the covenant channel: covenant violations are followed by lower credit-line availability, using firm fixed effects and lagged violation indicators.
- The results are most persuasive as evidence about the contractual conditions under which lines of credit are available.

5.2 Main threats and how the paper addresses them

- **Liquidity demand versus supply.** Firms with different liquidity demand might choose different liquidity mixes. The bank-liquidity-to-total-liquidity ratio partly addresses this by focusing on the composition of liquidity.
- **Cash denominator problem.** Scaling by total assets could mechanically relate low cash and line use. The paper scales key variables by non-cash assets.
- **Distress heterogeneity.** Cash flow matters most among high-distress firms, consistent with bank supply and covenant concerns rather than only borrower preference.
- **Covenant reporting.** Covenant violations are observed only when disclosed in filings, so the violation sample may miss some renegotiated or unreported violations.
- **Search classification error.** The full-sample line indicator is based on text search; the hand-collected random sample is used to validate and enrich the classification.

Interpretation: The empirical design is best read as a careful measurement and mechanism study, rather than as a clean natural experiment.

6 Tables and figures

6.1 Table 1: Summary statistics

Read:

- The full sample has 4,503 firms and 28,447 firm-year observations; the random sample has 300 firms and 1,908 firm-year observations.
- In the random sample, 74.8% of firm-years have a line of credit.
- Total lines average 15.9% of assets; unused lines average 10.2% of assets; used lines average 5.7% of assets.
- Total line liquidity accounts for about half of total liquidity when cash and lines are combined.
- Covenant violations occur in 8.0% of random-sample firm-years.

Economic message: Lines of credit are not a marginal financing product. They are large, common, and central to public firms' liquidity management.

6.2 Table 2: Which firms utilize bank lines of credit?

Read:

- Lines are common across industries, but the incidence varies: services have lower use, while wholesale and retail trade have very high use.
- Firms with S&P credit ratings use lines heavily; 94.5% of rated firm-years have a line.
- Even among firm-years with no debt outstanding, 31.9% have a line of credit.
- Conditional on having a line, the used share is meaningful, especially for firms with debt outstanding.

Economic message: Lines of credit are not only for opaque or unrated firms. They remain important even for firms with public debt access and for firms that appear debt-free on the balance sheet.

Table 1
Summary statistics.

Summary statistics						
	Full sample				Random sample	
Variable	Mean	Median	St. Dev.	Variable	Mean	Median
<i>Line of credit variables</i>						
Has line of credit (0,1)	0.817	1.000	0.387	Has line of credit (0,1)	0.748	1.000
				Total line of credit/assets	0.159	0.112
				Unused line of credit/assets	0.102	0.069
				Used line of credit/assets	0.057	0.000
				Total line/(total line + cash)	0.512	0.569
				Unused line/(unused line + cash)	0.450	0.455
				Violation of financial covenant (0,1)	0.080	0.000
<i>Firm characteristics</i>						
Book debt/assets	0.204	0.171	0.19	Book debt/assets	0.205	0.169
EBITDA/assets – cash	0.026	0.125	0.358	EBITDA/assets – cash	0.034	0.126
Tangible assets/assets – cash	0.340	0.277	0.239	Tangible assets/assets – cash	0.331	0.275
Net worth, cash adjusted	0.456	0.445	0.238	Net worth, cash adjusted	0.451	0.459
Assets – cash	1.608	102	11,434	Assets – cash	1.441	116
Market-to-book, cash adjusted	2.944	1.532	3.469	Market-to-book, cash adjusted	2.790	1.498
Industry sales volatility	0.045	0.035	0.033	Industry sales volatility	0.045	0.036
Cash-flow volatility	0.091	0.052	0.109	Cash-flow volatility	0.091	0.052
Not in an S&P index (0,1)	0.696	1.000	0.46	Not in an S&P index (0,1)	0.684	1.000
Traded over the counter (0,1)	0.125	0.000	0.328	Traded over the counter (0,1)	0.142	0.000
Firm age (years since IPO)	15	10	13	Firm age (years since IPO)	14	9

This table presents summary statistics for two samples of non-financial firms from 1996 through 2003. The left panel describes the full sample (28,447 firm-year observations), and the right panel describes the random sample of 300 firms (1908 firm-year observations). *Market-to-book*, is the market value of assets less cash divided by the book value of assets less cash balances. *Net worth, cash adjusted*, is the net worth balances divided by book assets less cash.

(a) Table 1: Summary statistics

Table 2
Which firms utilize bank lines of credit?

			Conditional on having line of credit		
	Line of credit (0,1)	Some debt (0,1)	Debt/assets	Total line/assets	Used line/total line
<i>Industry</i>					
Agriculture, minerals, construction	0.761	0.862	0.303	0.306	0.490
Manufacturing	0.731	0.809	0.205	0.216	0.267
Transportation, communications, and utilities	0.860	0.907	0.366	0.163	0.338
Trade—wholesale	0.932	0.915	0.256	0.221	0.321
Trade—retail	0.920	0.938	0.286	0.194	0.283
Services	0.630	0.701	0.245	0.204	0.304
<i>Corporate credit rating</i>					
No S&P corporate credit rating	0.697	0.765	0.196	0.220	0.319
S&P corporate credit rating	0.945	0.995	0.384	0.190	0.245
<i>Debt outstanding</i>					
No debt outstanding	0.319	0.000	0.000	0.110	0.000
Debt outstanding	0.847	1.000	0.267	0.221	0.325

This table presents data on the use of lines of credit by firms in the random sample of 300 firms (1908 firm-year observations). It reports cell means for subsamples by industry, by having a corporate credit

(b) Table 2: Which firms utilize bank lines of credit?

6.3 Table 3: Bank lines of credit and firm characteristics

Read:

- Cash flow positively predicts whether a firm has a line of credit in both the full and random samples.
- Cash flow also positively predicts the share of liquidity supplied by total or unused lines of credit.
- Firm size is strongly positive: larger firms are more likely to obtain and use lines.
- Market-to-book is negative, consistent with high-growth firms relying less on lines or holding more cash.
- Industry sales volatility is positive, consistent with lines being useful for working capital and seasonal liquidity needs.

Economic message: The key fact is that firms with high cash flow use more bank liquidity. This is the opposite of a simple view in which weaker firms should demand more backup liquidity from banks.

6.4 Figure 1: Use of line of credit versus cash holdings across cash-flow distribution

Read:

- The figure plots cash/assets and the fraction of firms with a line of credit by cash-flow decile.
- Moving from the lowest cash-flow decile to the middle of the distribution, line use rises sharply and cash holdings fall sharply.
- At very high cash-flow deciles, cash rises again and line use falls somewhat, reflecting high-growth/high-market-to-book firms such as cash-rich technology firms.

Economic message: The visual pattern captures the central substitution margin: low-cash-flow firms hold cash because bank liquidity is less available to them.

Table 3
Bank lines of credit and firm characteristics.

Dependent variable Regression type	Firm has line of credit [0,1] Probit (marginal effects)		Total line/(total line + cash) OLS		Unused line/(unused line + cash) OLS	
Sample	Full (1)	Random (2)	Random (3)	With line of credit (4)	Random (5)	With line of credit (6)
[EBITDA/ assets – cash] _{t-1}	0.107** (0.011)	0.124* (0.059)	0.109** (0.041)	0.086 (0.056)	0.085* (0.038)	0.120* (0.056)
[Tangible assets/ assets – cash] _{t-1}	0.015 (0.019)	0.070 (0.084)	0.041 (0.072)	0.014 (0.064)	0.025 (0.068)	-0.006 (0.068)
[Ln(assets – cash)] _{t-1}	0.040** (0.003)	0.052** (0.015)	0.054** (0.010)	0.032** (0.009)	0.054** (0.010)	0.041* (0.009)
[Net worth, cash adjusted] _{t-1}	-0.094** (0.016)	-0.037 (0.075)	-0.106 (0.059)	-0.136** (0.052)	-0.053 (0.054)	-0.040 (0.054)
[Market-to-book, cash adjusted] _{t-1}	-0.013** (0.001)	-0.022** (0.006)	-0.031** (0.004)	-0.037** (0.005)	-0.025** (0.004)	-0.030** (0.006)
[Industry sales volatility] _{t-1}	0.674** (0.193)	2.571** (0.885)	0.989** (0.382)	0.081 (0.319)	1.069** (0.401)	0.354 (0.366)
[Cash-flow volatility] _{t-1}	-0.020 (0.035)	-0.201 (0.146)	-0.383** (0.125)	-0.374* (0.177)	-0.291* (0.118)	-0.271 (0.181)
[Not in an S&P index[0, 1]]	0.035* (0.013)	0.046 (0.059)	0.065 (0.039)	0.047 (0.034)	0.034 (0.039)	0.026 (0.035)
[Traded over the counter[0, 1]]	-0.014 (0.013)	-0.045 (0.061)	0.077 (0.047)	0.129** (0.036)	0.022 (0.043)	0.029 (0.044)
Ln[Firm age (years since IPO)] _{t-1}	-0.003 (0.005)	-0.005 (0.024)	-0.027 (0.018)	-0.034* (0.016)	-0.012 (0.018)	-0.021 (0.016)
Number of observations	28,447	1908	1908	1428	1908	1428
Number of firms	4503	300	300	255	300	255
R ²	0.20	0.27	0.39	0.31	0.36	0.24

This table presents coefficient estimates from regressions relating the use of a line of credit to various lagged firm characteristics. Columns 1 and 2 report the estimated marginal effects (or effect of going from 0 to 1 for indicator variables) of lagged firm characteristics on the probability of having a line of credit from maximum likelihood probit estimation using the full and random sample, respectively. Columns 3 through 6 report the coefficient estimates from an OLS estimation using the random sample; the estimation relates two different measures of the bank liquidity to total liquidity ratio to lagged firm characteristics. The estimation reported in columns 4 and 6 isolates the intensive margin of the bank liquidity to total liquidity ratio by focusing only on firms that have a line of credit. Regressions include year and 1-digit industry indicator variables; standard errors are clustered at the firm level.

**, * statistically distinct from 0 at the 1 and 5% level, respectively.

(a) Table 3: Lines of credit and firm characteristics

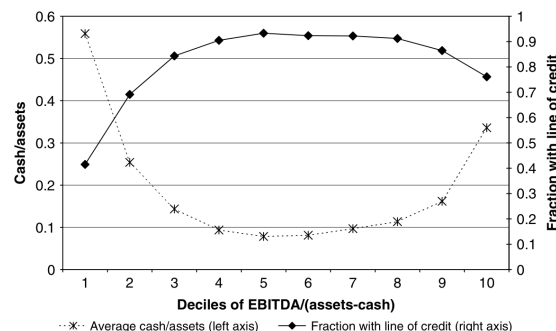


Figure 1
Use of line of credit versus cash holdings across cash-flow distribution

The figure maps the mean cash to total assets ratio and the fraction of firms that have a line of credit across cash-flow deciles. Going from left to right is going from firms in the lowest cash-flow decile to firms in the highest cash-flow decile.

(b) Figure 1: Lines of credit versus cash across cash-flow distribution

6.5 Table 4: Bank lines of credit and financial distress likelihood

Read:

- Cash flow is positively related to line use among firms with high distress likelihood.
- The interaction between cash flow and low distress is negative.
- This means that cash flow is much less important for safer firms.
- Low-distress firms are more likely to have lines and more unused line liquidity, but the cash-flow gradient is concentrated among riskier firms.

Economic message: The bank-liquidity channel is state contingent. Cash flow matters most when lenders are most concerned about repayment and covenant compliance.

6.6 Table 5: Financial covenants from LPC's Dealscan

Read:

- About 71.6% of credit-line loans in the Dealscan sample contain financial covenants.
- Coverage covenants are the most common, appearing in 70.0% of loans with covenant information.
- Debt-to-cash-flow covenants appear in 48.5%, and net-worth covenants appear in 46.1%.
- A substantial share of credit lines have cash-flow-based covenants.

Economic message: This table gives the contractual foundation for the paper. Bank liquidity is contingent on accounting performance, especially cash-flow performance.

Table 4
Bank lines of credit and financial distress likelihood.

Dependent Variable Regression type Sample	Firm has line of credit {0,1} Probit (marginal effects) Full (1)	Unused line/unused line + cash) OLS Random (2)	Unused line/unused line + cash) OLS Random (3)
[EBITDA/(assets – cash)] _{t-1}	0.116** (0.012)	0.130* (0.062)	0.081* (0.039)
[Low distress likelihood {0,1}] _{t-1}	0.089** (0.017)	0.165** (0.055)	0.191** (0.039)
[EBITDA/(assets – cash)] _{t-1} *	-0.294** (0.074)	-0.447* (0.221)	-0.452** (0.148)
[Low distress likelihood {0,1}] _{t-1}	0.029 (0.019)	0.109 (0.082)	0.064 (0.067)
[Tangible assets/(assets – cash)] _{t-1}	0.040** (0.003)	0.052** (0.014)	0.056** (0.009)
[Ln(assets – cash)] _{t-1}	-0.094** (0.016)	-0.064 (0.074)	-0.083 (0.054)
[Net worth, cash adjusted] _{t-1}	-0.011** (0.001)	-0.019** (0.005)	-0.021** (0.004)
[Market-to-book, cash adjusted] _{t-1}	0.606** (0.190)	2.410** (0.868)	0.888* (0.393)
[Industry sales volatility] _{t-1}	0.008 (0.035)	-0.152 (0.143)	-0.225* (0.112)
[Cash-flow volatility] _{t-1}	0.008 (0.035)	-0.152 (0.143)	-0.225* (0.112)
Number of observations	28,447	1908	1908
Number of firms	4503	300	300
R ²	0.20	0.27	0.38

This table presents coefficient estimates from regressions relating the use of a line of credit to various lagged firm characteristics, with a particular focus on how the likelihood of financial distress affects the use of lines of credit. *Low distress likelihood* is an indicator variable that is 0 if the firm has an *Altman's z-score* below the sample median and 1 if the firm has an *Altman's z-score* above the sample median. Regressions include year and 1-digit industry indicator variables, and information asymmetry control variables in Table 3; standard errors are clustered at the firm level. **,* statistically distinct from 0 at the 1 and 5% level, respectively.

Table 5
Financial covenants, from LPC's Dealscan.

Loans with financial covenant {0,1}	0.716
Conditional on having financial covenant, loans with:	
Coverage covenant {0,1}	0.700
Debt to cash flow covenant {0,1}	0.485
Net worth covenant {0,1}	0.461
Debt to net worth {0,1}	0.130
Current ratio covenant {0,1}	0.114
Leverage ratio covenant {0,1}	0.201
No cash flow-based covenant {0,1}	0.249

This table presents data from LPC's Dealscan on the use of covenants among public corporations. The total sample includes 11,758 loan deals by 4011 public firms. *Coverage covenant* includes fixed charge coverage, interest coverage, cash interest coverage, and debt service coverage covenants. *Net worth covenant* includes net worth and tangible net worth covenants.

(b) Table 5: Financial covenants from Dealscan

(a) Table 4: Lines of credit and financial distress likelihood

6.7 Table 6: Causes and consequences of financial covenant violations

Read:

- Lower EBITDA/assets strongly predicts covenant violations.
- Other accounting variables have weaker explanatory power than cash flow.
- A lagged covenant violation reduces total line availability and unused line availability.
- A violation also reduces bank liquidity as a share of total liquidity.

Economic message: This is the mechanism table. Cash-flow deterioration leads to covenant violations, and covenant violations lead banks to restrict access to liquidity.

6.8 Figure 2: Effect of a covenant violation on availability of line of credit

Read:

- The used portion of the line rises before the covenant violation.
- Total and unused line availability fall after the violation.
- The figure shows an economically large decline in available bank liquidity following covenant default.

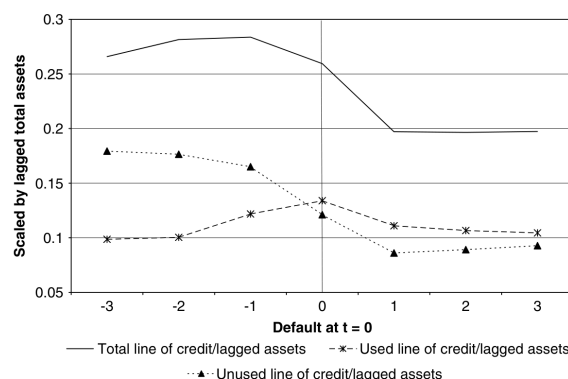
Economic message: Credit lines are not unconditional insurance. When the firm gets close to financial trouble, the bank can restrict the buffer.

Table 6
The causes and consequences of financial covenant violations.

	Covenant violation _t {0,1} (1)	Total line _t / assets _{t-1} (3)	Unused line _t / assets _{t-1} (4)	[Total line/ [(Total line + cash)] _t (5)	[Unused line [(unused line + cash)] _t (6)
EBITDA _t /assets _{t-1}	-0.577** (0.105)	-0.487** (0.112)			
[Debt/assets] _t		0.592* (0.246)			
[Net worth/assets] _t		-0.007 (0.242)			
Current ratio _t		-0.007 (0.010)			
Market-to-book ratio _t		-0.003 (0.008)			
Ln(total assets) _t		0.033 (0.029)			
Covenant violation _{t-1} {0,1}		-0.041* (0.018)	-0.036** (0.011)	-0.070** (0.027)	-0.122** (0.038)
EBITDA _{t-1} /assets _{t-2}		0.071 (0.046)	0.062 (0.039)	0.073 (0.054)	0.089 (0.066)
[Debt/assets] _{t-1}		0.005 (0.164)	-0.012 (0.109)	0.381* (0.193)	0.177 (0.213)
[Net worth/assets] _{t-1}		0.014 (0.141)	0.016 (0.095)	0.257 (0.169)	0.152 (0.174)
Current ratio _{t-1}		-0.001 (0.004)	0.002 (0.004)	-0.022** (0.008)	-0.018* (0.008)
Market-to-book ratio _{t-1}		0.010* (0.005)	0.002 (0.004)	-0.001 (0.005)	-0.010 (0.006)
Ln(total assets) _{t-1}		-0.116** (0.025)	-0.087** (0.018)	0.008 (0.025)	-0.016 (0.030)
Number of observations	1428	1387	1174	1174	1174
Number of firms	255	249	246	246	246
R ²	0.21	0.24	0.61	0.57	0.65

Columns 1 and 2 present regression coefficients from firm fixed effects regressions relating the probability of a covenant violation to firm characteristics. Columns 3 through 6 present regression coefficients from firm fixed effects regressions relating line of credit balances to a covenant violation in the previous year. The sample used for columns 1 and 2 includes only firms that have a line of credit. The sample used for columns 3 through 6 includes only firms that have a line of credit in the previous year. Regressions include year indicator variables, and standard errors are clustered at the firm level.

**, * statistically distinct from 0 at the 1 and 5% level, respectively.



(b) Figure 2: Covenant violation and line availability

(a) Table 6: Covenant violations

6.9 Table 7: Correlation with other measures of financial constraints

Read:

- The line-of-credit constraint measure is positively correlated with traditional measures but far from identical to them.
- Its correlation with the payout measure is 0.09, with size 0.19, with bond rating 0.18, and with commercial paper rating 0.26.

- Traditional measures are more highly correlated with each other than with the line-of-credit measure.

Economic message: Credit-line access captures a distinct dimension of financial constraints, closer to actual liquidity insurance than to broad observable firm type.

6.10 Table 8: Availability of bank lines of credit and cash-flow sensitivity of cash

Read:

- Firms constrained by the line-of-credit measure save cash out of cash flow; unconstrained firms do not.
- Among firms classified as constrained by payout, size, bond rating, or commercial paper rating, the positive cash-flow sensitivity of cash is driven by firms without credit-line access.
- Traditional constraint measures have explanatory power, but line access refines and strengthens them.

Economic message: The paper converts a contract feature into an empirical measure of constraints. Access to a line of credit is a direct measure of whether a firm can rely on external liquidity rather than hoarding cash.

Table 7
Correlation with other measures of financial constraints.

	Line of credit	Payout decile	Size decile	Bond rating
Payout ratio decile	0.09			
Size decile	0.19	0.44		
Bond rating	0.18	0.29	0.68	
CP rating	0.26	0.32	0.48	0.52

This table presents correlations between various measures of financial constraints used in the literature. *Line of credit* takes on the value 1 if the firm (i) has access to a line of credit in every year of the sample, and (ii) maintains cash flows above the median firm in every year of the sample. *Bond rating* and *commercial paper rating* take on the value 1 if the firm ever has an S&P corporate credit rating and commercial paper rating through the sample, respectively.

Note: All correlations are statistically distinct from 0 at the 1 percent level.

(a) Table 7: Correlation with other financial-constraints measures

Table 8
Availability of bank lines of credit and the cash-flow sensitivity of cash.

Dependent variable Δ Cash Holdings	Panel A			
	(1)	(2)	(3)	(4)
	Splitting constrained firms by:			
		Low cash flow, no line of credit	Low cash flow, line of credit	High cash flow, no line of credit
1. Lines of credit				
Unconstrained	-0.062 (0.045)			
Constrained	0.084 ^{*,+} (0.010)	0.085 ^{**} (0.010)	0.063 ^{**} (0.016)	0.028 (0.097)
	Panel B			
	(1)	(2)	(3)	
			Splitting constrained firms by:	
			Constrained, by line of credit measure	Unconstrained, by line of credit measure
2. Payout ratio				
Highest 3 deciles	0.110 ^{**} (0.025)			
Lowest 3 deciles	0.089 ^{**} (0.014)	0.090 ^{**} (0.014)		-0.077 (0.121)
3. Firm size (assets)				
Largest 5 deciles	-0.023 (0.027)			
Smallest 5 deciles	0.096 ^{*,+} (0.012)	0.096 ^{**} (0.012)		0.014 (0.083)
4. Bond rating				
Has a rating	0.016 (0.027)			
Does not have a rating	0.086 ^{*,+} (0.011)	0.089 ^{**} (0.011)		-0.042 (0.063)
5. Commercial paper rating				
Has a rating	-0.047 (0.047)			
Does not have a rating	0.082 ^{*,+} (0.010)	0.085 ^{**} (0.010)		-0.050 (0.052)

This table presents coefficient estimates from regressions relating the change in cash holdings to cash flow. The estimation follows that of Almeida et al. (2004), which they describe in their Equation (8) and Table 3. Each reported coefficient is the effect of cash flow on cash holdings from a separate regression. Panel A splits the sample on the basis of the line of credit measure of financial constraints in which firms that (i) have access to a line of credit in every year of the sample, and (ii) maintain cash flows above the median firm in every year of the sample are considered unconstrained. Panel B examines the measures of financial constraints used in Almeida et al. (2004). All estimations include year and firm fixed effects. Standard errors are heteroskedasticity-robust, clustered at the firm.

^{***} distinct from 0 at 1 and 5%, respectively; ⁺ distinct from unconstrained sample at 10% or better.

(b) Table 8: Bank lines and cash-flow sensitivity of cash

7 Short summary

- The paper studies why firms use bank lines of credit versus cash for liquidity management.
- The data are built from 10-K filings, including a full text-search sample and a hand-collected random sample.
- Lines of credit are large and common: in the random sample, around three-quarters of firm-years have a line, and total lines average roughly 16% of assets.
- High cash flow predicts both access to a credit line and a greater share of liquidity supplied by credit lines.
- Low-cash-flow firms hold more cash because lines are less reliable for them.
- The main mechanism is financial covenants: cash-flow declines predict covenant violations, and covenant violations lead banks to reduce line availability.
- Credit lines therefore are not fully committed liquidity insurance.
- Lack of access to a line of credit is a powerful measure of financial constraints.
- Traditional constrained-firm classifications matter mainly when they also identify firms without line access.

8 Conclusion

8.1 Core conclusion

The paper’s core conclusion is that lines of credit are conditional liquidity. They provide an important liquidity source for healthy firms, but they are not a complete substitute for cash because their availability depends on ongoing firm performance and covenant compliance.

8.2 Economic intuition

Banks are willing to provide revolving credit facilities when firms maintain sufficient cash flow. For strong firms, a line of credit is valuable because it provides flexible liquidity at low carrying cost. For weak firms, the line is less reliable because covenant violations can reduce access. Therefore, low-cash-flow firms rationally hold more cash.

8.3 Incremental contribution revisited

The paper advances corporate liquidity research by moving from the question “why do firms hold cash?” to “when can firms rely on external liquidity instead of cash?” Its answer is contractual: bank liquidity is supplied under covenants, and those covenants make credit-line access state-contingent.

8.4 Implications for financial constraints

The findings suggest that empirical work should treat credit-line access as a central measure of financial flexibility. A firm without a credit line is more likely to save cash out of cash flow, even if traditional proxies alone do not fully identify it as constrained.

8.5 Limitations

The evidence is observational, and credit-line access reflects both demand and supply. The full-sample line indicator is based on text search, and covenant violations may be incompletely reported. The paper nevertheless provides strong descriptive and mechanism-based evidence because it combines 10-K line data, hand-collected balances, covenant information, and cash policy tests.

8.6 Takeaway

Bank lines of credit are valuable but fragile liquidity. Firms with strong cash flow can use bank liquidity; firms with weak cash flow need cash because bank liquidity can contract precisely when it becomes most valuable.